Non-Inverting Lever Compliant Mechanism

Adapted as a Multi-Material Universal Gripper

Kunal Ganesh Kongari | Sr. No.: 26732 | MN 203 — DfAM Homework 1 | IISc Bengaluru | MTech Smart Manufacturing

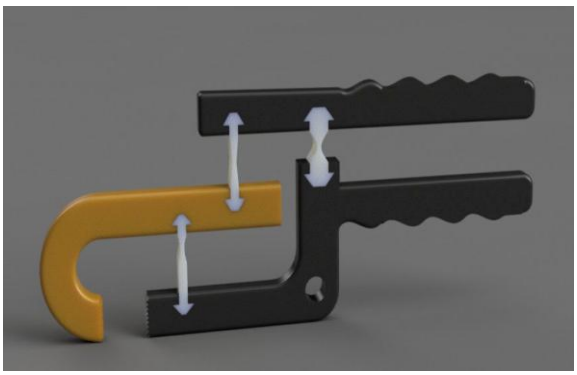

Fig. 1: CAD Render — PLA (black/amber) + TPU (white)

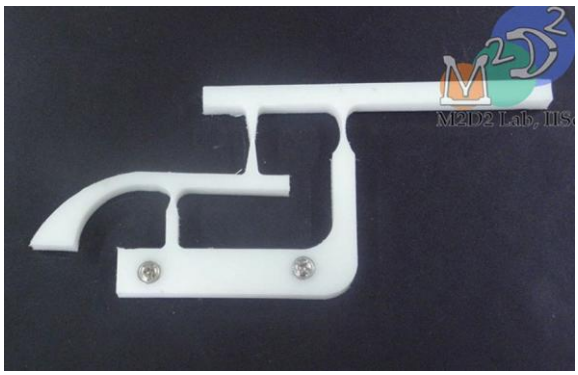

Fig. 2: Physical prototype from M2D2 Lab, IISc

1. Mechanism Selection and Function

The selected mechanism from the IISc-M2D2-CM collection is the Non-Inverting Lever Compliant Mechanism. Its defining characteristic is counter-intuitive, co-directional motion: pressing the right-side handle downward causes the left-side output arm to also move downward — opposite to what a conventional rigid lever would produce. This non-inverting behaviour is achieved entirely through the elastic deformation of slender flexure joints, with no rigid pin joints, bearings, or sliding contacts. The mechanism was originally conceived for miniaturized triggering applications such as dot-matrix printers, where its scalability and monolithic construction offer significant advantages.

Key functional advantages inherent to this compliant topology include: zero mechanical backlash, no assembly of discrete joint parts, passive return force stored as elastic energy, and straightforward scalability to different size ranges. Its non-intuitive motion is not a limitation but the core design intent — both input and output travel in the same direction, making it ideal for symmetric jaw-closing mechanisms where balanced grip force on an object is desired.

2. Design Concept: Universal Compliant Gripper

2.1 Identified Need

A common workshop and desk accessory need is a lightweight, tool-free gripper capable of securely holding objects of widely varying size, shape, and weight — such as coins, padlocks, tape rolls, and electrical chargers — for tasks like hands-free photography, inspection, or temporary fixturing. Existing rigid clamps are bulky, lack adaptability, and risk scratching delicate surfaces. A compliant gripper printed entirely from standard FDM materials addresses all these shortcomings with no moving parts beyond the flexures themselves.

2.2 AfAM Design Decisions

The non-inverting lever topology was adapted into a two-jaw gripper product following the AfAM (Adapt for Additive Manufacturing) philosophy. The following design decisions were made:

- **Function mapping:** The lever's co-directional output was utilized such that squeezing the ergonomic handle closes the gripper jaw symmetrically, providing balanced grip force on the held object.
- **Ergonomics:** The handle was sculpted with a palm-conforming curved profile and serrated grip ridges on the upper jaw, enabling comfortable single-hand actuation. The lower jaw was given a rounded hook profile for gentle contact with objects of different geometries.
- **Multi-material construction:** 6 total parts — 3 in rigid PLA (frame, upper jaw, lower jaw body) and 3 in flexible white TPU (compliant hourglass flexure joints). PLA provides structural stiffness; TPU

provides the elastic deformation required for compliant motion and the passive return force that holds jaws open at rest.

- **Weight reduction:** Handle bodies were hollowed internally with Mesh like structure, reducing material use while retaining structural stiffness under typical grip loads.
- **Aesthetics:** The design in the CAD render uses matte black PLA for the structural frame and amber/golden PLA for the moving jaw body, creating a bold two-tone scheme that also visually differentiates rigid and moving parts.

3. Fabrication and Testing

3.1 3D-Printed Prototype

The physical prototype shown in Figs. 3–8 was fabricated using white PLA and TPU on an FDM printer at IISc. The six parts were assembled by hand using dovetail joints, eliminating the need for any fasteners or adhesives. The assembled prototype clearly demonstrates the intended mechanism geometry, jaw actuation, ergonomic handle design, and full-scale gripping functionality.

Fig. 3: Hand scale & ergonomics

Fig. 4: Gripping a 5-rupee coin

Fig. 5: Holding a padlock (open)

Fig. 6: Squeezing to grip the padlock

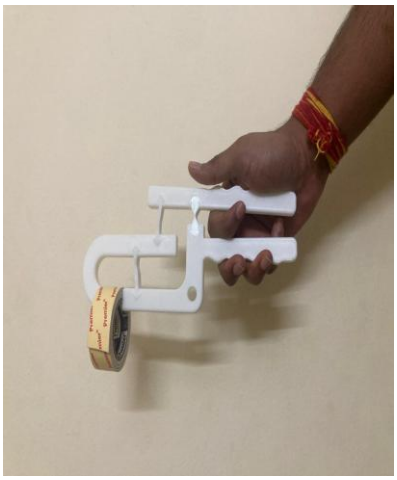

Fig. 7: Gripping a tape roll

Fig. 8: Holding an electrical charger

3.2 Functional Testing Results

The assembled gripper was tested on four everyday objects of varying geometry, size, and weight to validate adaptability:

- 2-rupee coin (~5 g, thin flat disc): Held stably between the serrated lower jaw tooth and curved upper hook. The flexures provided just enough preload force without bending the coin.
- Padlock (~150 g, cylindrical metal body): The rounded jaw profile accommodated the cylindrical body. The gripper held the padlock suspended in air, demonstrating meaningful load capacity for a purely compliant FDM part.
- Tape roll (~40 g, toroidal): The wide jaw opening accepted the tape roll; symmetric grip force centered it without deforming the roll.
- Electrical charger (~100 g, rectangular): The flat rectangular body was gripped across its narrower dimension; the serrated jaw surface prevented slipping even when tilted.

In all test cases, the gripper remained fully open at rest due to TPU flexure return force, closed smoothly with a single index-finger push, and released immediately upon removing finger pressure — confirming the mechanism's passive restoring behavior and the effectiveness of the multi-material compliant design approach.

4. Conclusion

This project demonstrates the AfAM approach by successfully transforming the abstract Non-Inverting Lever compliant mechanism topology into a functional, ergonomic, and versatile everyday product. The multi-material design (PLA + TPU) directly exploits additive manufacturing's capability to create parts with locally tuned stiffness in a single monolithic build cycle, without any separate hinge hardware or post-processing. The resulting universal gripper is lightweight, tool-free, surface-friendly, and capable of holding objects from a thin coin to a heavy padlock — validating the practical value of compliant mechanism design principles when thoughtfully adapted for 3D printing.